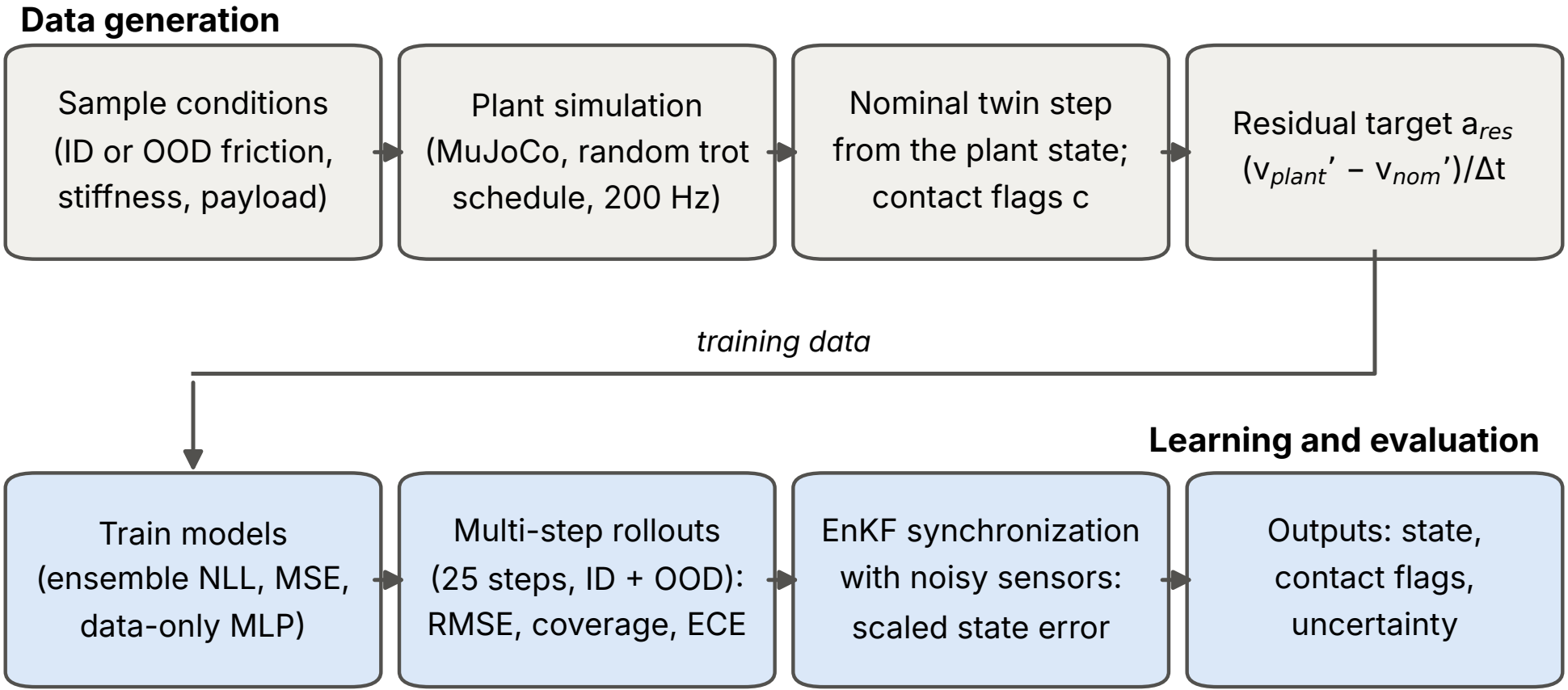
Data generation
Sample conditions (ID or OOD friction, stiffness, payload)
Plant simulation (MuJoCo, random trot schedule, 200 Hz)
Nominal twin step from the plant state; contact flags c
Residual target a_{res} $(v_{plant}' - v_{nom}')/\Delta t$
training data
Learning and evaluation
Train models (ensemble NLL, MSE, data-only MLP)
Multi-step rollouts (25 steps, ID + OOD): RMSE, coverage, ECE
EnKF synchronization with noisy sensors: scaled state error
Outputs: state, contact flags, uncertainty